IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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A presentation for the IBM Applied Data Science Capstone Course

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

• This project **analyzes the factors influencing successful recovery of the SpaceX Falcon 9 first stage** — a key innovation that has reduced launch costs and transformed the commercial space sector

• **Data** was **collected** from the **SpaceX public REST API** and **web-scraped historical launch records**, resulting in a unified, cleaned dataset of Falcon 9 launches prior to 2021

• **Exploratory data analysis (EDA)** using Python (Pandas, NumPy), SQL, geospatial mapping (Folium), and interactive dashboards (Plotly Dash) revealed:

➪ Launch success rates have improved over time, with certain launch sites (notably KSC LC-39A) and orbit types (ES-L1, GEO, HEO, SSO) showing higher recovery rates;

➪ Payload mass and launch site characteristics were analyzed for their impact on outcomes;

➪ Predictive modeling with four machine learning classifiers (Logistic Regression, SVM, Decision Tree, KNN) achieved up to 85% accuracy, with the Decision Tree model performing best;

➪ The models demonstrated high recall (100%) for successful launches, indicating strong predictive power for identifying successful recoveries

• Accurate forecasting of first-stage recovery enables more precise launch cost estimation and supports strategic pricing and operational decisions, strengthening Space X's competitive position in the orbital launch market

Introduction

•**Industry Context**: The commercial space sector is rapidly evolving, with companies like SpaceX leading innovation through reusable rocket technology, significantly reducing launch costs

•**Strategic Importance**: SpaceX's ability to recover and reuse the Falcon 9 first stage allows it to offer launches at $62 million—far below the industry average of $165 million—creating a competitive advantage

•**Technical Focus**: This report investigates the conditions under which the Falcon 9 first stage is successfully recovered, using publicly available launch data and machine learning techniques

•**Analytical Approach**: We collected and processed launch records via SpaceX's REST API and web scraping, performed exploratory data analysis and SQL-based insights, and built predictive models to assess landing outcomes

•**Business Impact**: By accurately predicting first-stage recovery, we can estimate launch costs more precisely, inform strategic pricing decisions, and strengthen our competitive positioning in the orbital launch market

Section 1

Methodology

Methodology

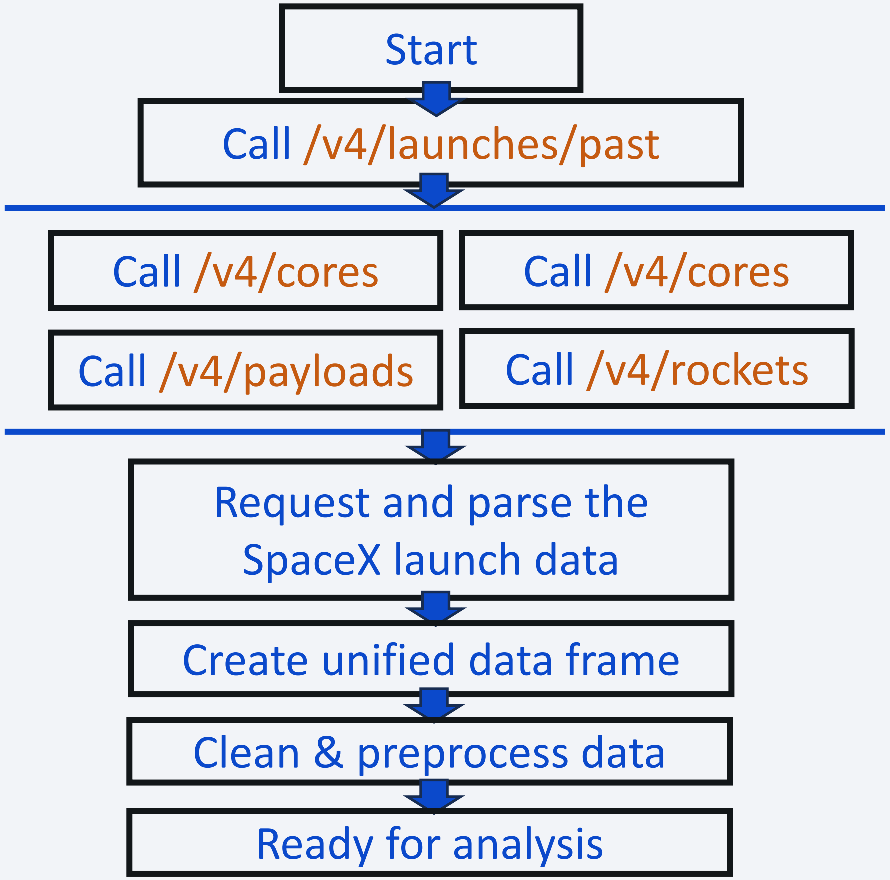
Executive Summary

- Data collection methodology:
  - Data was collected from SpaceX REST API and Web scraping of Wikipedia historical launch data; date ranges were conserved to keep consistency
- Perform data wrangling:
  - Create a “Class” label for use in data analysis and ML classification algorithms that classifies launches as successful or unsuccessful based on whether the first stage landed successfully
- Perform exploratory data analysis (EDA) using visualization (Pandas and NumPy) and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Four classification models were evaluated for their accuracy in predicting the success of the launch outcome

Data Collection – SpaceX REST API

SpaceX Public REST API was used to retrieve structured data about past Falcon 9 launches, including:

- General past launch data (/v4/launches/past)
- Rocket specifications, namely Booster Name and Version (/v4/rockets)
- Payload mass and orbit (/v4/payloads)
- Launch site name and coordinates (/v4/launchpads)
- Landing type and outcome (v4/cores)
- Core information (v4/cores)



Key API Endpoints

Full data collection using SpaceX REST API can be found here:
[IBM-Data-Science-CapStone/Data_Collection_SpaceX_REST_API.ipynb at main · adelinaivanova14/IBM-Data-Science-CapStone]

Data Collection – SpaceX REST API Data Cleaning

Cleaning steps included:

- Removing rows with multiple cores – rows including falcon rockets with 2 extra rocket boosters and rows that have multiple payloads in a single rocket
- Filtering the unified data frame to include only Falcon 9 launches
- Extract the date without the time & restrict the dates of the launches to pre-2021
- Replace missing values for payload mass with the mean value across the dataset (5 values replaced)

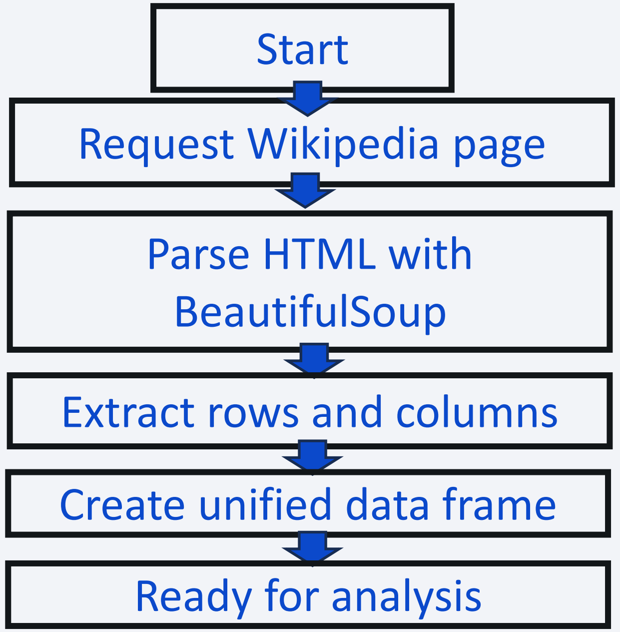
Final data frame contains 90 rows (instances) x 17 columns

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs
4	1	2010-06-04	Falcon 9	NaN	LEO	CCSFS SLC 40	None None	1	False	False	False
5	2	2012-05-22	Falcon 9	525.0	LEO	CCSFS SLC 40	None None	1	False	False	False
6	3	2013-03-01	Falcon 9	677.0	ISS	CCSFS SLC 40	None None	1	False	False	False
7	4	2013-09-29	Falcon 9	500.0	PO	VAFB SLC 4E	False Ocean	1	False	False	False
8	5	2013-12-03	Falcon 9	3170.0	GTO	CCSFS SLC 40	None None	1	False	False	False
...	...	...	...	...	...	...	...	...	...	...	...
89	86	2020-09-03	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS	2	True	True	True
90	87	2020-10-06	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS	3	True	True	True
91	88	2020-10-18	Falcon 9	15600.0	VLEO	KSC LC 39A	True ASDS	6	True	True	True
92	89	2020-10-24	Falcon 9	15600.0	VLEO	CCSFS SLC 40	True ASDS	3	True	True	True
93	90	2020-11-05	Falcon 9	3681.0	MEO	CCSFS SLC 40	True ASDS	1	True	False	True

Data Collection – Web Scraping of Historical Launch Records

To supplement the API data, Falcon 9 launch records from Wikipedia were scraped using BeautifulSoup with the following considerations:

- Data was from a snapshot of the Wikipage ''List of Falcon 9 and Falcon Heavy Launches'' updated on 9th June 2021
- Parsed values were transformed into a data frame



Final data frame contains 121 rows (instances) x 11 columns

	Flight No.	Launch site	Payload	Payload mass	Orbit	Customer	Launch outcome	Version Booster
0	1	CCAFS	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success\n	F9 v1.0B0003.1
1	2	CCAFS	Dragon	0	LEO	NASA	Success	F9 v1.0B0004.1
2	3	CCAFS	Dragon	525 kg	LEO	NASA	Success	F9 v1.0B0005.1
3	4	CCAFS	SpaceX CRS-1	4,700 kg	LEO	NASA	Success\n	F9 v1.0B0006.1
4	5	CCAFS	SpaceX CRS-2	4,877 kg	LEO	NASA	Success\n	F9 v1.0B0007.1

Full data collection using Web scraping can be found here:
[IBM-Data-Science-CapStone/Data_Collection_WebScraping.ipynb at main · adelinaivanova14/IBM-Data-Science-CapStone]

Data Wrangling

Goal: unify mission outcome labels (i.e. landing success) into Training Labels for ML models, where 1 signifies that the booster landed successfully and 0 means it was unsuccessful

Start → Exploratory Data Analysis (EDA) → Creating outcome labels

Exploratory Data Analysis (EDA)

- Identify data type (numerical vs categorical) per column
- Determine the number of launches per site
- Determine the orbit the probe reached
- Determine the number and occurrence of landing outcomes

Creating outcome labels

- Group outcomes where the second stage did not land successfully
- Create a ''Class'' label where the value (1/0) signifies the success of the first stage landing

Full data wrangling workflow can be found here:
IBM-Data-Science-CapStone/Data_wrangling.ipynb at main · adelinaivanova14/IBM-Data-Science-CapStone

EDA with Data Visualization

Plotted charts and rationale:

Chart	Rationale
Flight number vs. Payload mass with overlayed landing outcome	Explore how flight number (a surrogate for time & learnings) and payload mass affect landing outcome
Flight number vs. Launch site with overlayed landing outcome	Explore how flight number (a surrogate for time & learnings) and launch site affect landing outcome
Payload mass vs. Launch site with overlayed landing outcome	Explore if certain launch sites have restrictions in payload mass and how that impacts landing outcome
Landing success rate vs. Orbit type	Identify what are the optimal orbits that the launch should reach to provide the best chance for successful landing outcome
Flight number vs. Orbit type with overlayed landing outcome	Explore if there is a correlation between the flight number and orbit, and the success of the landing outcome
Payload mass vs. Orbit type with overlayed landing outcome	Explore how payload mass affects landing success rate per orbit
Launch success vs. year	Identify the trend in launch success with time (years)

Full EDA with pandas and NumPy workflow can be found here:

EDA with SQL

Database Setup

•Created a SQLite database from the cleaned Falcon 9 launch dataset
•Loaded the Pandas DataFrame into a SQL table using df.to_sql()
•Remove blank rows

Basic exploration queries

•SELECT distinct launch site names
•SELECT the total payload mass carried by boosters launched by NASA (CRS)
•SELECT the average payload mass carried by booster v. F9 v1.1

Landing outcomes exploration queries

•SELECT the date when the first successful landing outcome in ground pad was achieved
•SELECT all the booster versions that have carried the maximum payload mass
•SELECT and rank the list of landing outcomes from 2010 to 2017

Full EDA with SQL workflow can be found here:
[IBM-Data-Science-CapStone/Data_Analysis_EDA_SQL.ipynb at main · adelinaivanova14/IBM-Data-Science-CapStone]()

Build an Interactive Map with Folium

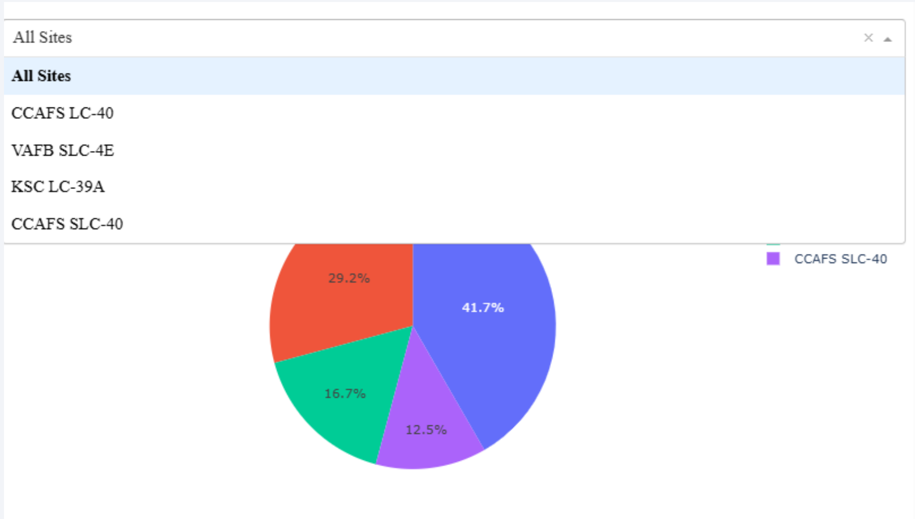
	Start	Map all launch sites	Mark launch success for each site	Calculate site distance to proximities
ACTION	Load SpaceX launch dataset with latitude and longitude data for each site	Create a circle and marker objects for each launch site	Create a marker for all launch records using MarkerCluster and colour the marker based on the success of the launch	Create a marker and polyline to the site's closest coastline, city, railway and highway
REASON		Identify the positioning of all launch sites to allow subsequent analysis of their strategic location in relation to success rates and proximities	Explore the relationship between site location and launch success	Explore how sites are positioned with respect to proximal key locations, what is desirable to have in proximity and what is avoided

Full Interactive map building with Folium workflow can be found here:
[IBM-Data-Science-CapStone/Data_Analysis_Folium.ipynb at main · adelinaivanova14/IBM-Data-Science-CapStone]

Build a Dashboard with Plotly Dash

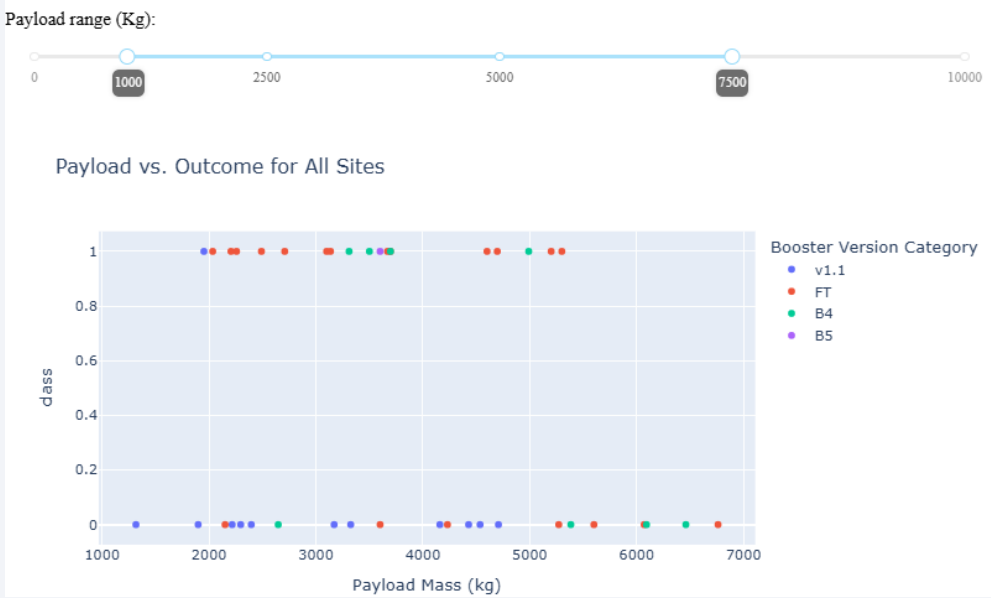
Section I:

Explore percentage of successful launches for all sites and per site in the form of interactive pie chart (callback) with drop-down input component **(dcc.Dropdown)**



Section II:

Explore if variable payload and booster version (callback) is correlated to mission outcome using a range slider to select payload (**dcc.RangeSlider**)

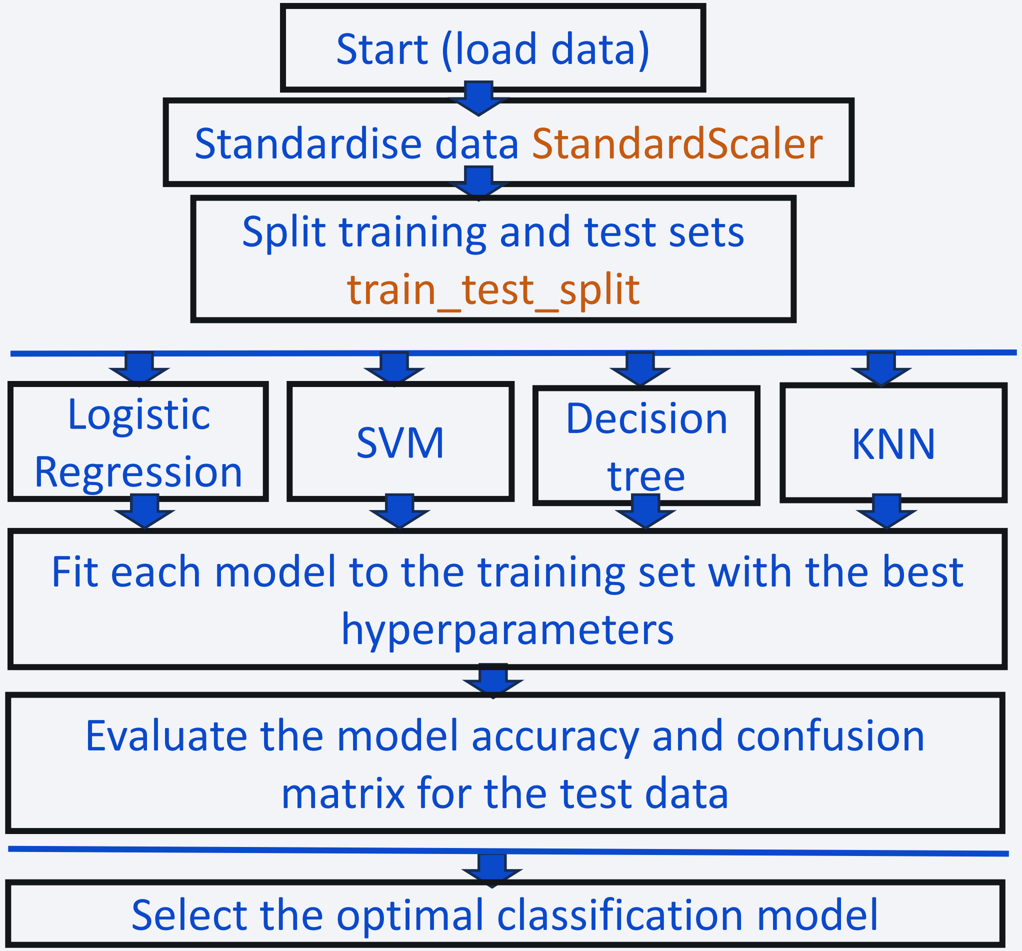


Full Dashboard with Plotly Dash workflow can be found here:
IBM-Data-Science-CapStone/Data_Analysis_Plotly_Dash.py at main · adelinaivanova14/IBM-Data-Science-CapStone

Predictive Analysis (Classification)

The Scikit-learn library was used to standardise the data and build classification models:

- Standardise data (StandardScaler)
- Split data into training and test sets with 20% test size (18 samples) and random state of 2 (train_test_split)
- Create and fit to training dataset ML classification models with a Grid Search cv=10 to optimise hyperparameters for models (GridSearchCV):
  - Logistic regression (LogisticRegression)
  - Support Vector Machine (SVM) (SVC)
  - Decision Tree (DecisionTreeClassifier)
  - K-Nearest Neighbour (KNeighborsClassifier)
- Evaluate the model accuracy and confusion matrix for the test data to select the optimal model (score & confusion_matrix)



Key Scikit-learn modules

Full Predictive Analysis workflow can be found here:
[IBM-Data-Science-CapStone/Predictive_Analysis_Classification.ipynb at main · adelinaivanova14/IBM-Data-Science-CapStone]

Results

4 sections

Visual Exploratory Data Analysis (EDA) for general insights

Geospatial launch site proximity analysis with Folium

Interactive dashboard analysis of launch success rates and relationship with relevant features

Predictive modelling of launch success using ML classifiers

Key points:

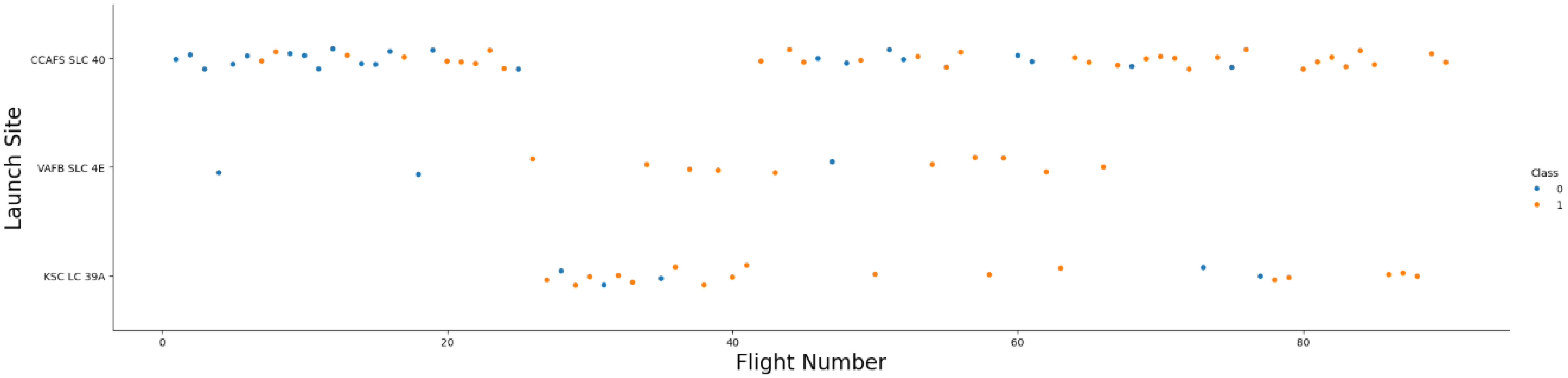
Launch success is a metric of whether the first stage landed successfully – our target parameter

The "Class" value represents launch success, where 0 is an unsuccessful launch and 1 is a successful launch

Section 2

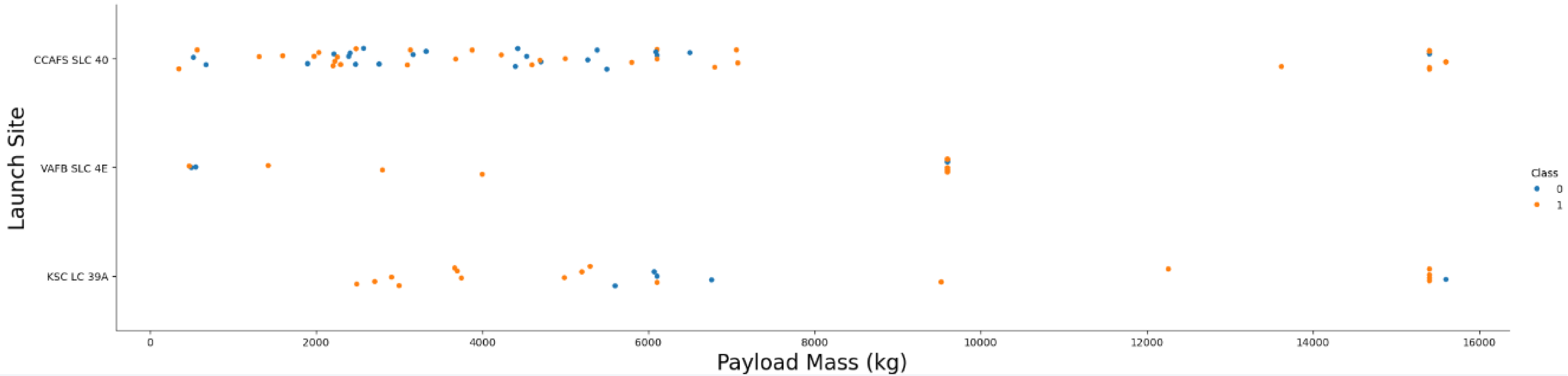
Insights drawn from EDA

Flight Number vs. Launch Site



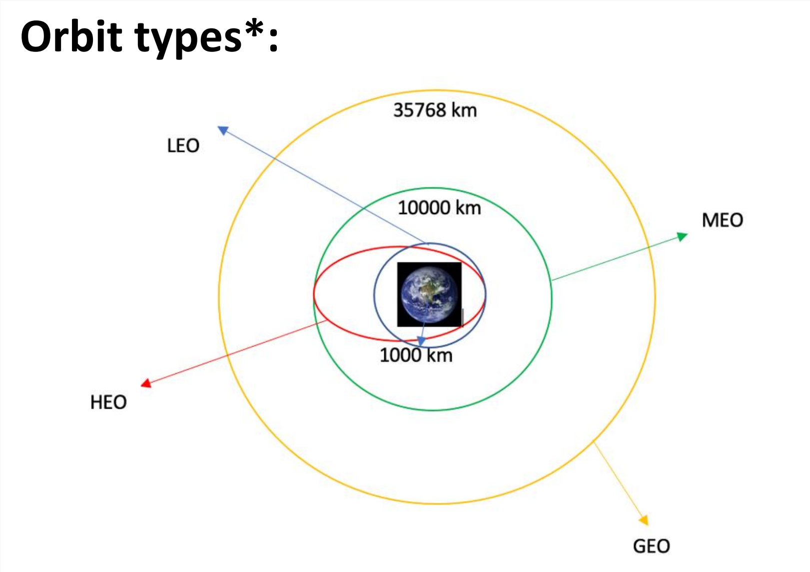
- Success rate for almost every launch site increases with flight number = appears logical as more experience acquired over time leads to improvements in landing outcomes
- CCAFS SLC 40 launch site handled early launches almost exclusively (like first commissioned launch site), with success rate in early flights being relatively low (<50%)

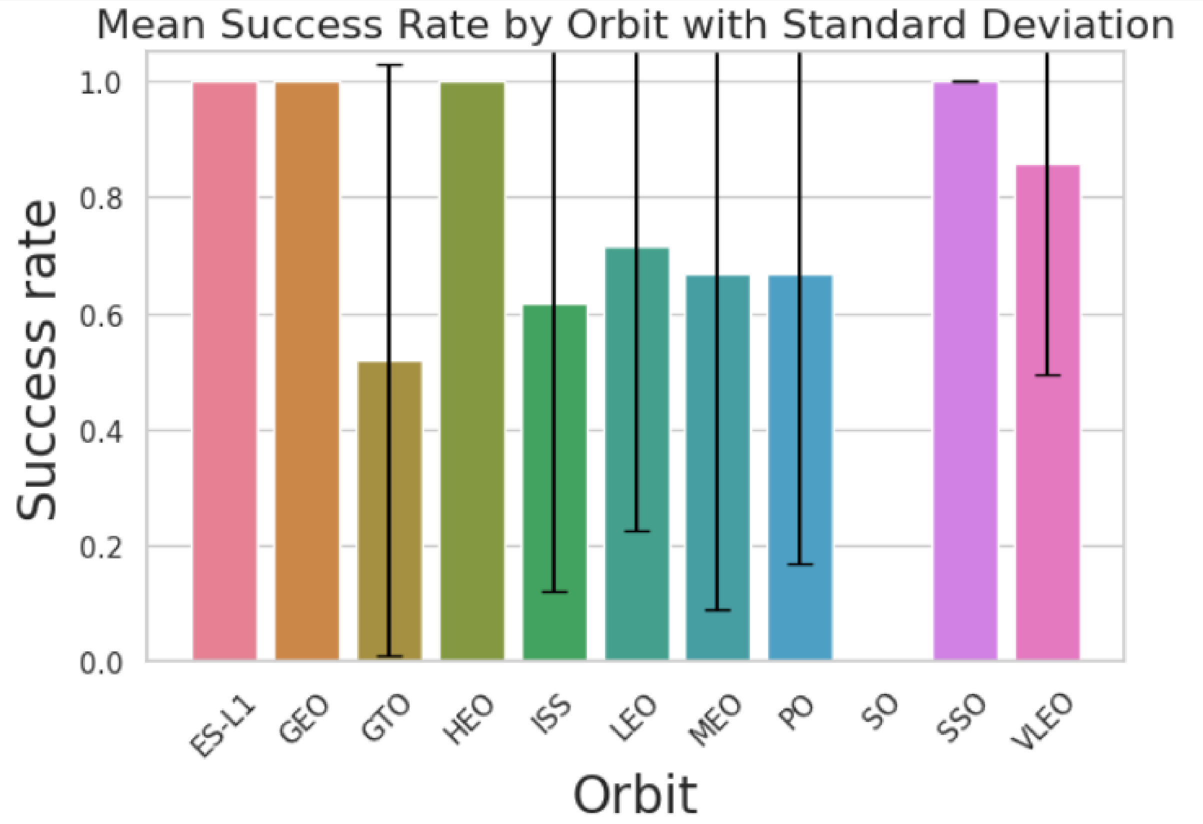
Payload vs. Launch Site



- Payload mass does not appear directly correlated to landing outcome
- For the VAFB-SLC launch site, there are no rockets launched for heavy payload mass (>10000 kg); this is likely suggesting a physical limitation in terms of rocket size for this launch site

Success Rate vs. Orbit Type

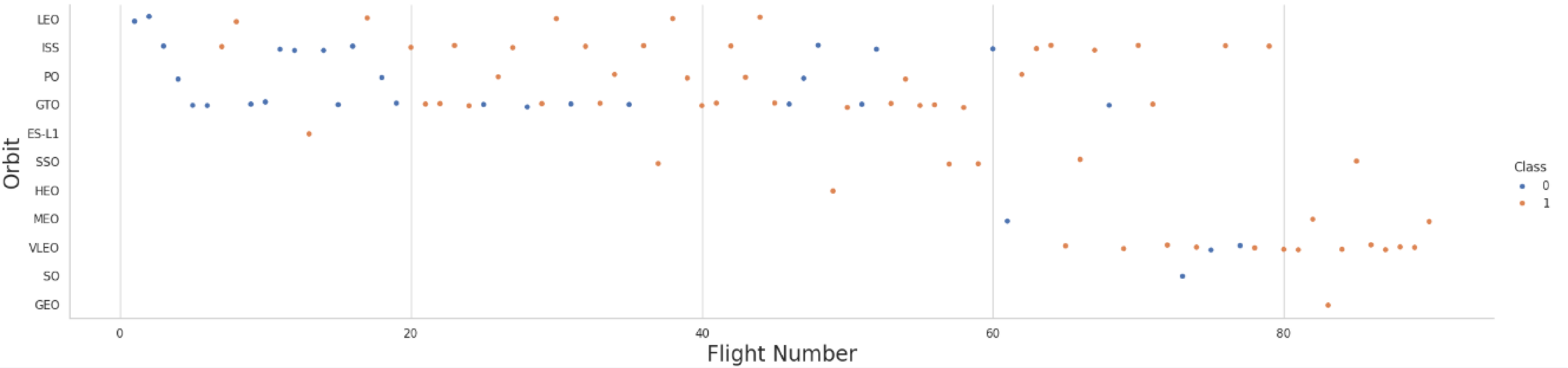




- Launches reaching ES-L1, GEO, HEO and SSO orbits have the highest success rates
- The other orbits results in a 50-60% success rate for successful landing outcome

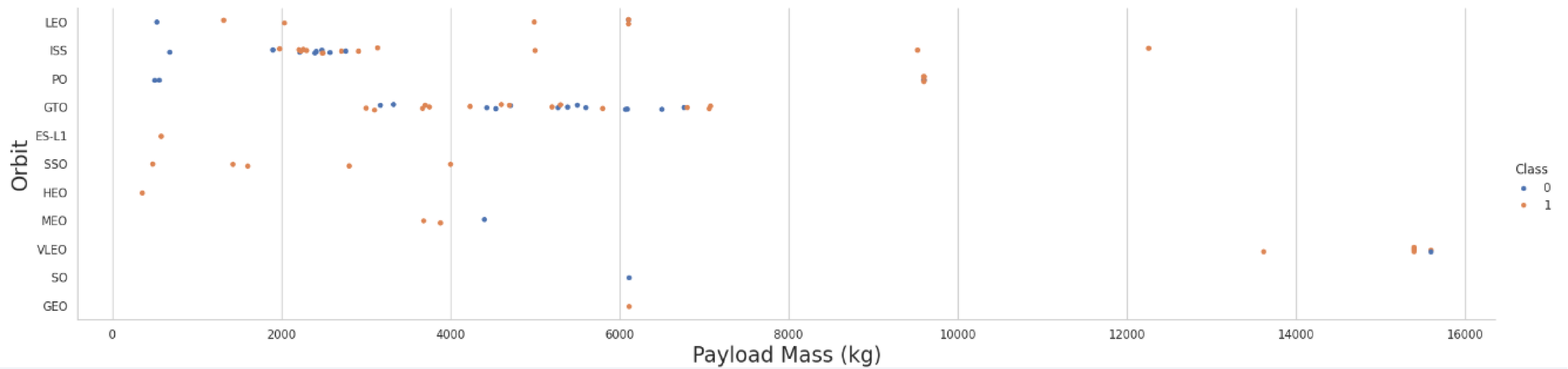
*Full explanation of the different orbit types can be found in the appendix

Flight Number vs. Orbit Type



- In the LEO orbit, success appears to be related to the number of flights
- In the GTO orbit, there appears to be no relationship between flight number and success

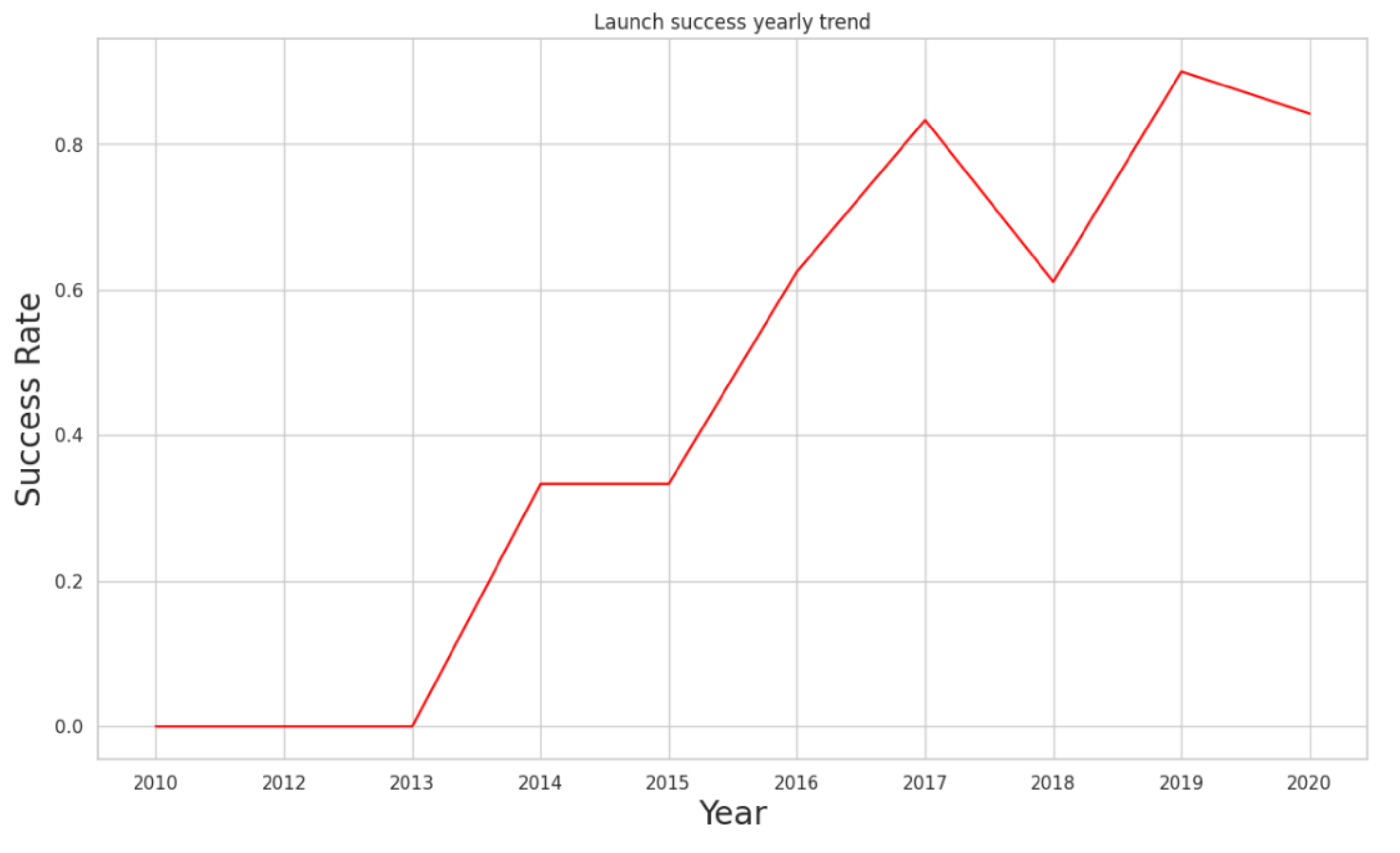
Payload vs. Orbit Type



- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS orbits
- In the GTO orbit, it is difficult to determine if payload mass affects landing success as both outcomes are present

Launch Success Yearly Trend

- Success rate has been steadily increasing since 2013 until 2017.
- A drop in success rate was observed for the period 2017-2018, but the upward trend was restored post-2018
- Some decrease in success rate surrounding the COVID-19 pandemic in 2020-2021 could be expected due to less staff, decreased operating capacity, etc,



Exploration of Launch Site Names

4 unique SpaceX launch sites were identified using SQL

- **VAFB SLC-4E**
- **CCAFS LC-40**
- **KSC LC-39A**
- **CCAFS SLC-40**

With this knowledge, we can identify records from distinct launch sites to inform us of specific launch site(s) landing success rates

Ex: 5 records where launch site name begins with CCA

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Exploration of Payload Mass Data

- The total payload mass carried by boosters launched by NASA (CRS) Customer – 45596 kg
- The average payload mass carried by booster version F9 v1.1 – 2928.4 kg

Boosters which have successfully landed on drone ship and had payload mass: 6000>mass>4000

Booster Version
F9 FT B1021.2
F9 FT B1022
F9 FT B1026
F9 FT B1031.2

Boosters that have carried the maximum payload mass

Booster Version	Booster Version
F9 B5 B1048.4	F9 B5 B1049.5
F9 B5 B1049.4	F9 B5 B1060.2
F9 B5 B1051.3	F9 B5 B1058.3
F9 B5 B1056.4	F9 B5 B1051.6
F9 B5 B1048.5	F9 B5 B1060.3
F9 B5 B1051.4	F9 B5 B1049.7

Key Values Calculated

Exploration of Mission and Landing Outcomes

- First successful landing outcome on ground pad – 22/12/2015

Landing outcomes
04/06/2010 – 20/03/2017

Landing outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Total number of successful and failure **mission** outcomes

Mission Outcome	Count
Success	100
Failure (in flight)	1

Only one mission was unsuccessful => high mission success rate is correlated with higher chance of the first stage successfully landing back to be reused

2015 **failed drone ship** Landing outcomes of the first stage

Booster version	Launch site
F9 v1.1 B1012	CCAFS LC-40
F9 v1.1 B1015	CCAFS LC-40

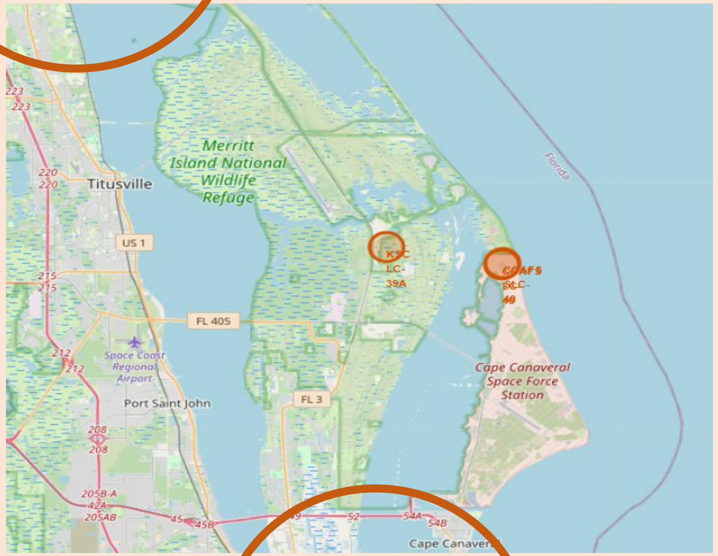
Same launch site for both failed drone ship landings

Section 3

Launch Sites Proximities Analysis

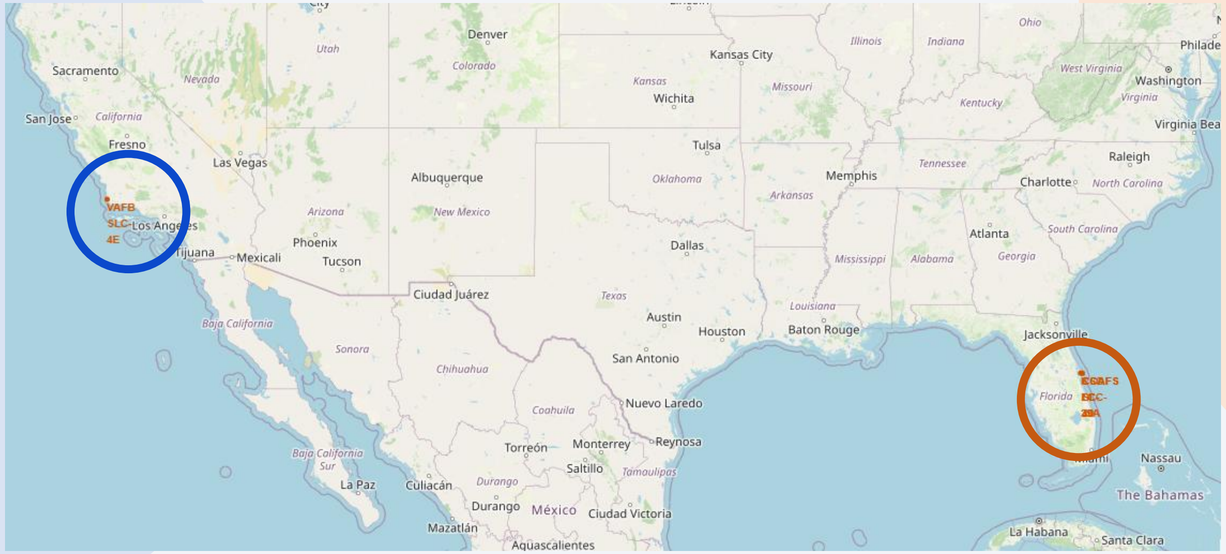
SpaceX Launch Site Locations

KSC LC-39A



VAFB SLC-4E

4 unique SpaceX launch sites near the coast, in the south part of the US - closer to Equator to shorten distance from Earth to outer orbits

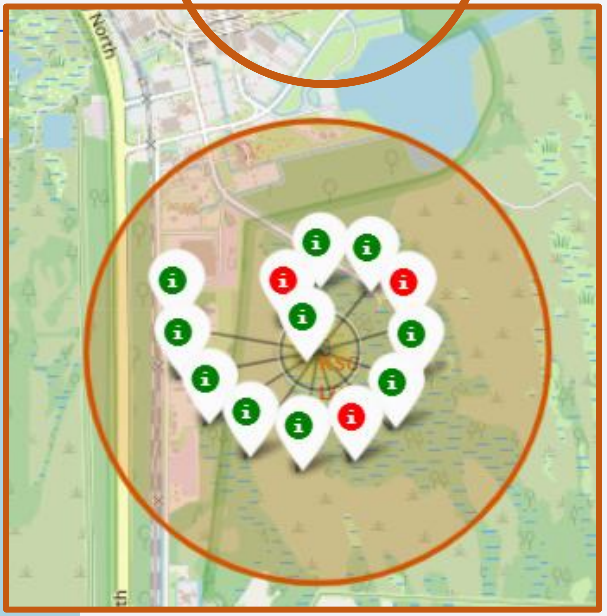


CCAFS LC-40

CCAFS SLC-40

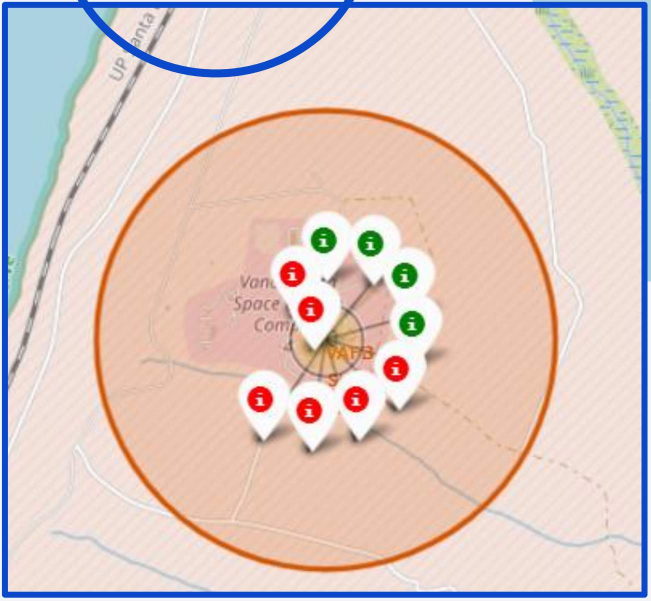
Launch Outcomes per Site

Highest success rate → KSC LC-39A

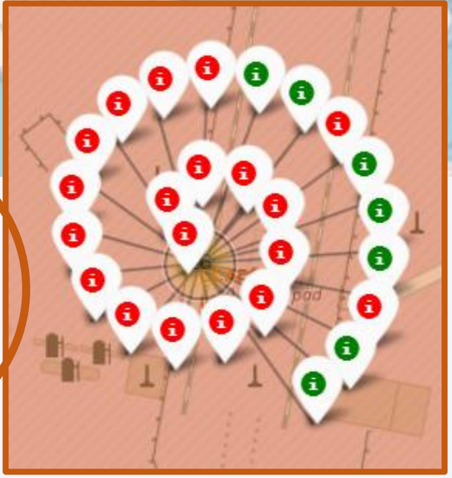


Launch outcome legend:
Green – successful
Red – unsuccessful

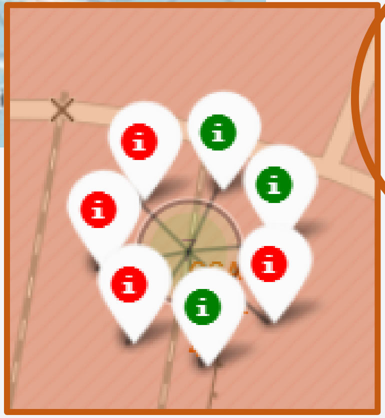
VAFB SLC-4E



CCAFS LC-40

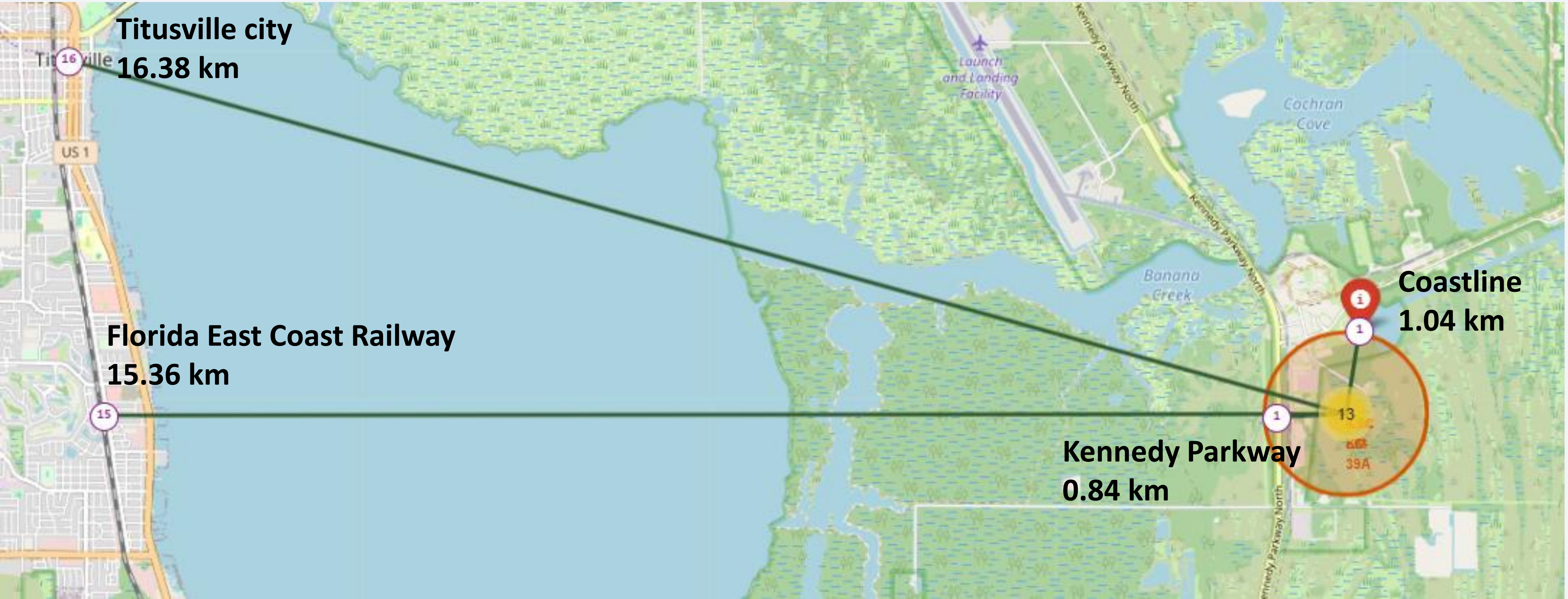


CCAFS SLC-40



Landmark Proximity Analysis for KSC LC-39A

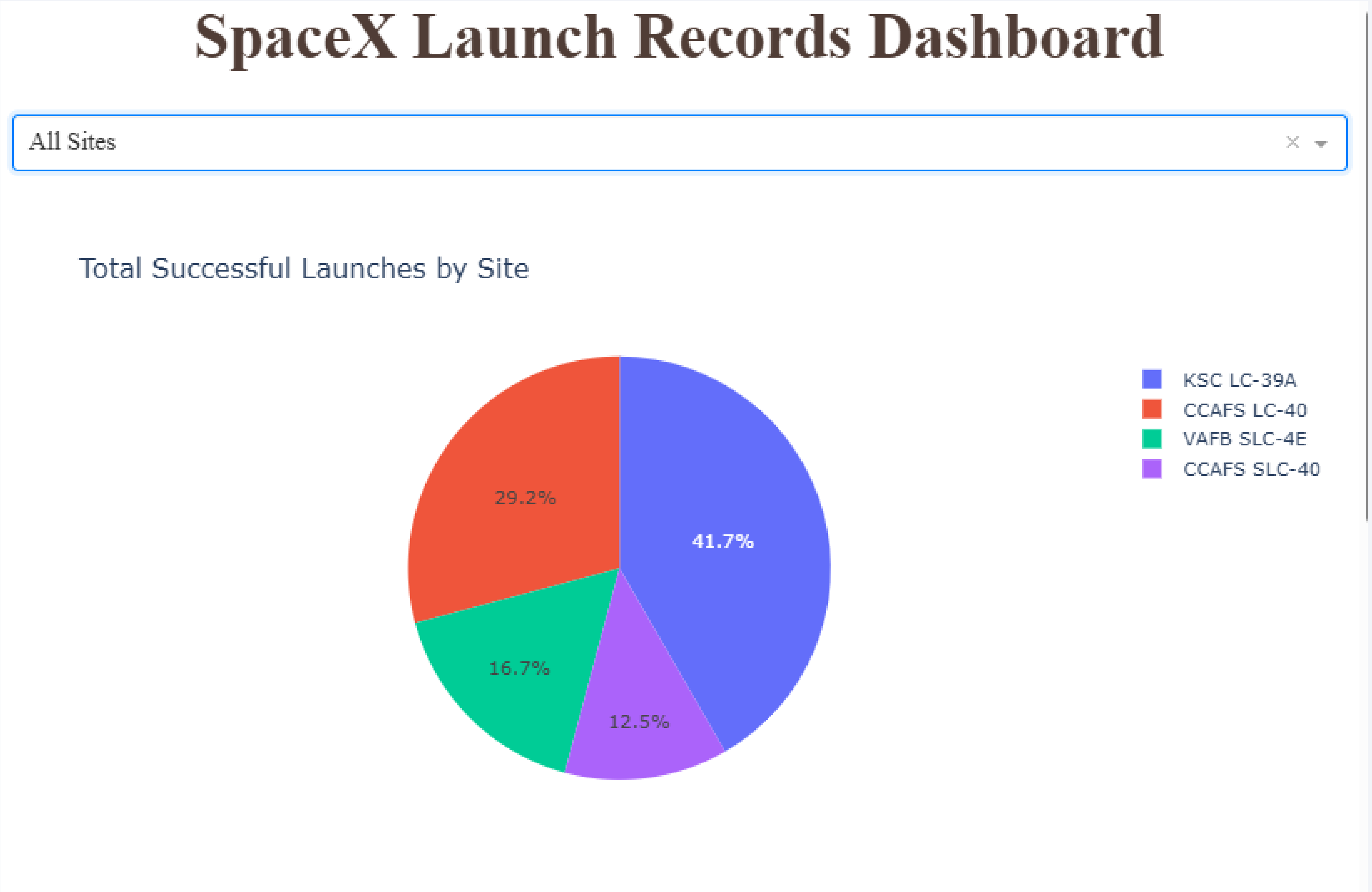
KSC LC-39A is the site with the **highest launch success rate** – it is located in close proximity to the coast and transportation (roads), but away from major cities and railways likely due to safety considerations in case of emergencies



Section 4

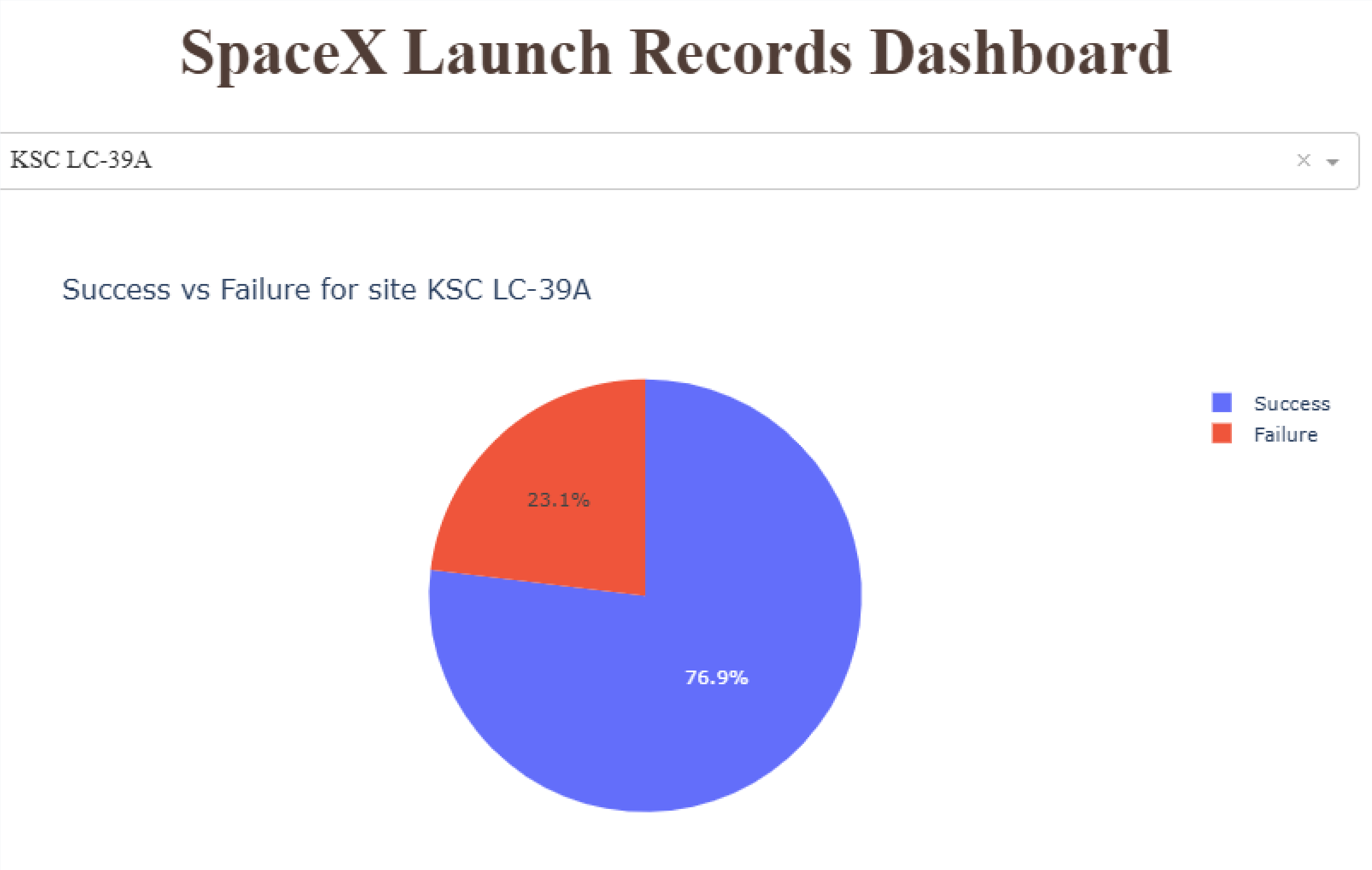
Build a Dashboard with Plotly Dash

Launch Success Count per Launch Site



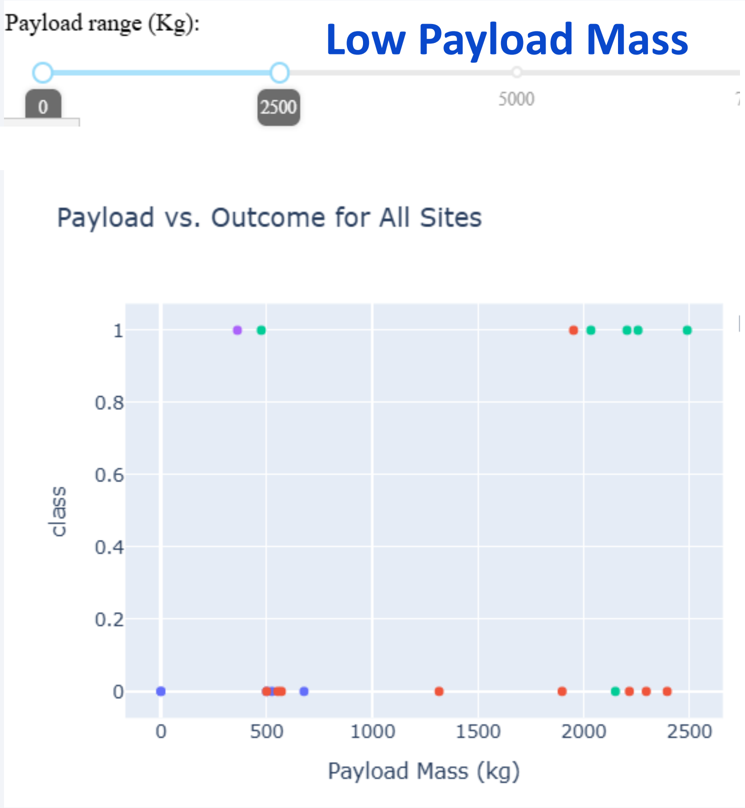
- Launch site with most successful launches – KSC LC-39A (as observed in geo-analysis with Folium)
- Launch site with least successful launches – CCAFS-SLC40 (but also site with lowest number of total launches)

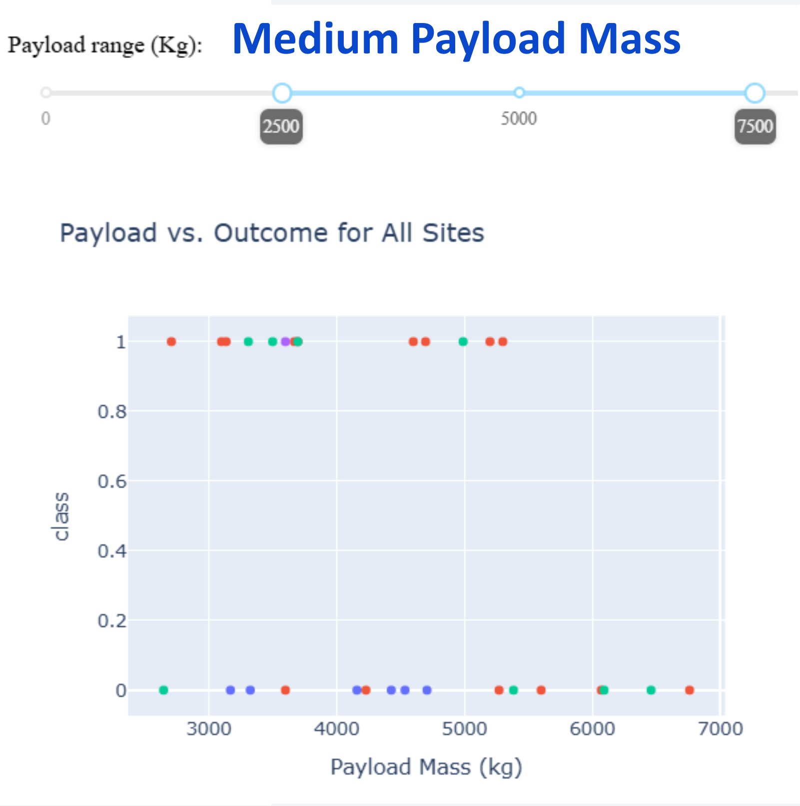
Successful vs Failed Launches from KSC LC-39A

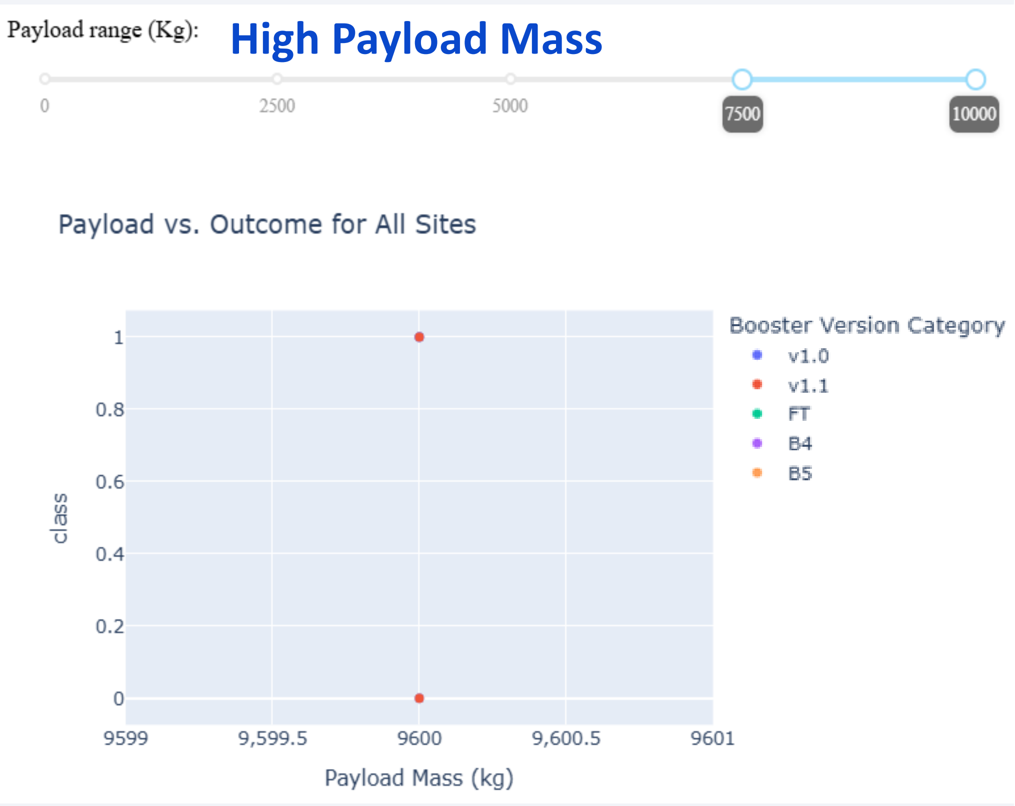


- >75% success rate for launches from this site

Payload mass vs Launch Outcome







With light payloads, booster B4 has the highest launch success rate, while booster FT is significantly underperforming

With medium payloads, boosters B4 and FT have the highest launch success rate, while booster v1.1 is underperforming

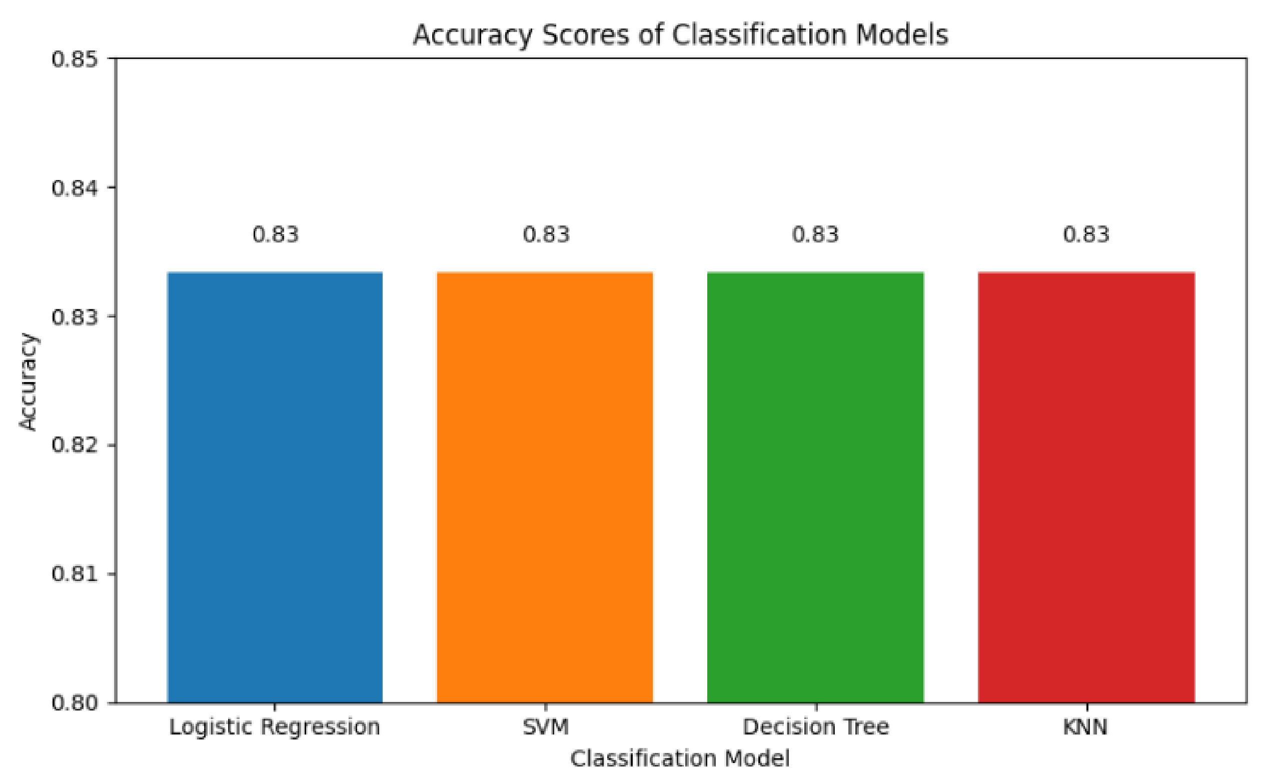
With heavy payloads, data is scarce. It appears only booster FT has been trialed

Section 5

Predictive Analysis (Classification)

Classification Accuracy

Although the four classification models had different accuracy on the training dataset, they displayed the same accuracy on the test dataset



! Accuracy is similar between training and test datasets (~85%), suggesting that we do not have overfitting

Since the Decision Tree model has the best GridSearchCV score, it was selected as the best-performing model

	Best scores
Logistic regresssion	0.846429
SVM	0.848214
Decision tree	0.891071
KNN	0.848214

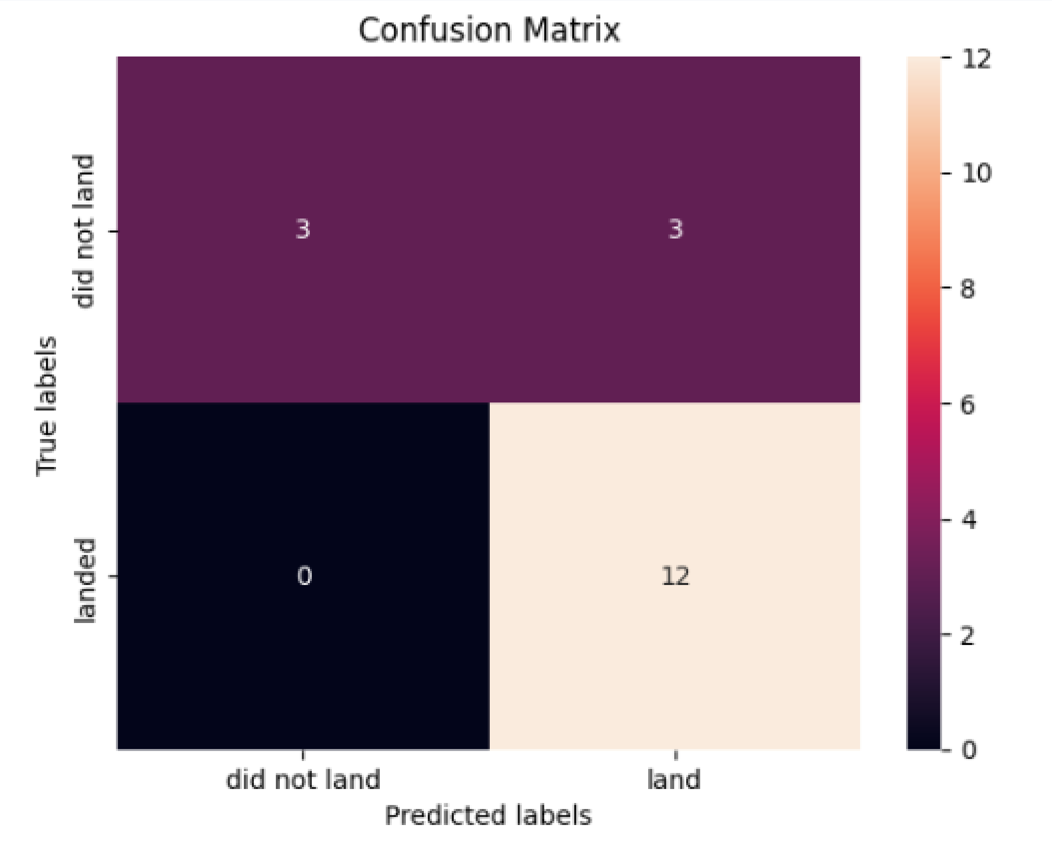
Confusion Matrix for Classification Models

- Key observations:
- All four models have the same Confusion Matrix
- The Confusion Matrix displays a recall of 1 (100%), so the models have a great success at identifying successful launches
- The precision of the models is 75%, and the F1 score is 86%

Rationale for the models making similar predictions:

1) The models are learning similar decision boundaries
2) Features in the dataset may thus be highly informative

⇨ Key consideration: the test dataset may be too small, we may consider exploring different train/test splits



Conclusions

Experience Improves Success Rates

The success rate for Falcon 9 first-stage recovery increases with flight number, indicating that operational experience and iterative improvements over time are major predictors of successful landings.

Launch Site Matters

Certain launch sites, especially KSC LC-39A, consistently achieve higher success rates. Proximity to the coast and transportation infrastructure, but distance from major cities and railways, appears to contribute to safer and more successful recoveries.

Orbit Type Influences Recovery

Launches targeting ES-L1, GEO, HEO, and SSO orbits have the highest first-stage recovery success rates. Orbit selection is thus a significant predictor of landing outcome.

Payload Mass Has Limited Direct Impact

While payload mass alone does not strongly correlate with landing outcome, certain sites (e.g., VAFB SLC-4E) have physical limitations that restrict heavy payload launches, indirectly affecting success rates.

Machine Learning Models Validate Feature Importance

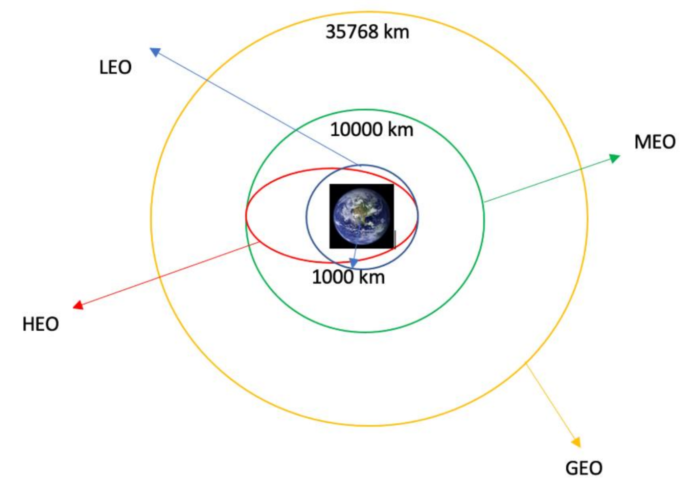
Predictive models (Decision Tree, SVM, Logistic Regression, KNN) achieved up to 85% accuracy, with high recall for successful launches. The models confirm that launch site, orbit type, and accumulated flight experience are the most informative features for predicting first-stage recovery.

Full GitHub Repository:
[adelinaivanova14/IBM-Data-Science-CapStone: CapStone project data and results for IBM Applied Data Science Course]

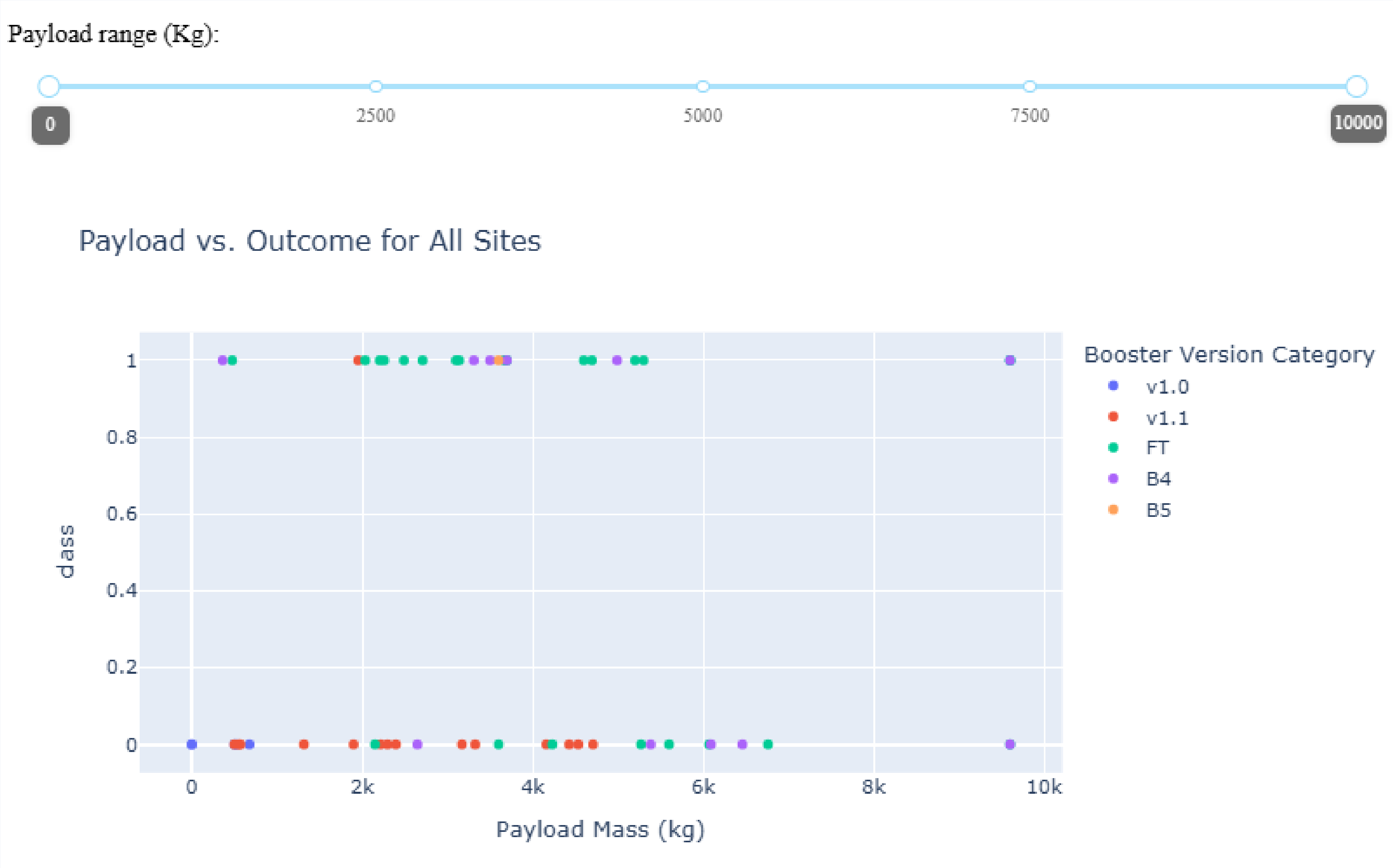
Appendix – Orbit types

- **LEO**: Low Earth orbit (LEO)is an Earth-centred orbit with an altitude of 2,000 km (1,200 mi) or less (approximately one-third of the radius of Earth),[1] or with at least 11.25 periods per day (an orbital period of 128 minutes or less) and an eccentricity less than 0.25.[2] Most of the manmade objects in outer space are in LEO [1].
- **VLEO**: Very Low Earth Orbits (VLEO) can be defined as the orbits with a mean altitude below 450 km. Operating in these orbits can provide a number of benefits to Earth observation spacecraft as the spacecraft operates closer to the observation[2].
- **GTO**(Geostationary Transfer Orbit): A geostationary transfer orbit is an elliptical Earth orbit used to transfer satellites from low Earth orbit (LEO) to geostationary orbit (GEO). In a GTO, the perigee (closest point to Earth) is much lower than GEO altitude, while the apogee (farthest point) reaches approximately 22,236 miles (35,786 kilometers) above Earth's equator — the altitude of a geostationary orbit. Satellites in GTO use onboard propulsion to circularize their orbit at GEO altitude, where they can provide services such as weather monitoring, communications, and surveillance. [3] .
- **SSO (or SO)**: It is a Sun-synchronous orbit also called a heliosynchronous orbit is a nearly polar orbit around a planet, in which the satellite passes over any given point of the planet's surface at the same local mean solar time [4] .
- **ES-L1** :At the Lagrange points the gravitational forces of the two large bodies cancel out in such a way that a small object placed in orbit there is in equilibrium relative to the center of mass of the large bodies. L1 is one such point between the sun and the earth [5] .
- **HEO** A highly elliptical orbit, is an elliptic orbit with high eccentricity, usually referring to one around Earth [6].
- **ISS** A modular space station (habitable artificial satellite) in low Earth orbit. It is a multinational collaborative project between five participating space agencies: NASA (United States), Roscosmos (Russia), JAXA (Japan), ESA (Europe), and CSA (Canada) [7]
- **MEO** Geocentric orbits ranging in altitude from 2,000 km (1,200 mi) to just below geosynchronous orbit at 35,786 kilometers (22,236 mi). Also known as an intermediate circular orbit. These are "most commonly at 20,200 kilometers (12,600 mi), or 20,650 kilometers (12,830 mi), with an orbital period of 12 hours [8]
- **HEO** Geocentric orbits above the altitude of geosynchronous orbit (35,786 km or 22,236 mi) [9]
- **GEO** It is a circular geosynchronous orbit 35,786 kilometres (22,236 miles) above Earth's equator and following the direction of Earth's rotation [10]
- **PO** It is one type of satellites in which a satellite passes above or nearly above both poles of the body being orbited (usually a planet such as the Earth [11]

Orbit types:



Appendix – Payload mass vs Launch Outcome for All Launch Sites



Thank you!